

New IoT-based Shared Parking Lock System

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Abstract— Parking spaces are in short supply in some urban core areas. Some residential communities have privately owned parking spaces that are often unused, leaving them unused and wasted. Currently, most parking space ground locks require manual opening and closing, making it inconvenient to drive in and out of the space, requiring you to get out of your car and manually open and close the ground lock.

We have designed a smart ground lock for this purpose. The owner can use their phone's Bluetooth to control the ground lock's opening and closing. If the parking space is unused for an extended period, the owner can share the space with their phone. The ground lock's screen will then display the owner's phone number, indicating that the space has been shared. This allows someone to call the number on the LCD screen to inquire about the time and cost of sharing the space. The owner will then provide a password. Entering the password unlocks the ground lock.

Keywords— parking space, smart parking space, shared parking space

1. Background

Parking spaces are in short supply in some urban core areas. Some residential communities have private parking spaces that are often unused, leaving them unused and wasted. Currently, most parking space ground locks require manual opening and closing, making it inconvenient to drive in and out of

the space, requiring drivers to get out of their cars and manually open and close the lock.

2. Research Overview

Research Step 1: Wireless control methods for ground locks, such as Bluetooth, Wiff, or Zigbee.

Research Step 2: Use actuators for opening and closing ground locks, such as servo motors, DC motors, electromagnets, or linear motors.

Research Step 3: Password input sensor selection: Choose a touch sensor, a non-contact method (without direct contact, the sensor can be sensed through a housing), or a photoelectric sensor, which is susceptible to light interference.

3. Research Process

3.1 Prototype Design

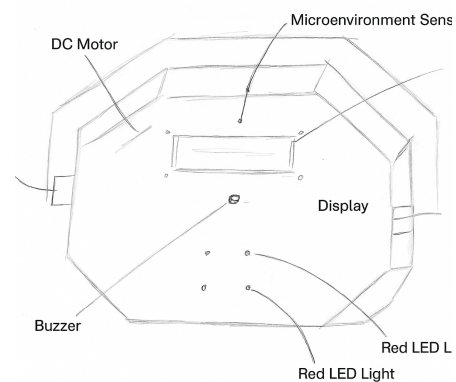


Figure 1 Prototype

3.2 Controller Parameters

- Advanced CGF control module, offering low power consumption and excellent compatibility.

- Operating voltage: 7.5-9 VDC.
- Two high-torque motor drive ports and two servo motor drive ports.
- Eight sensor inputs and multiple outputs for LEDs, LCDs, and more.
- Mini USB port.

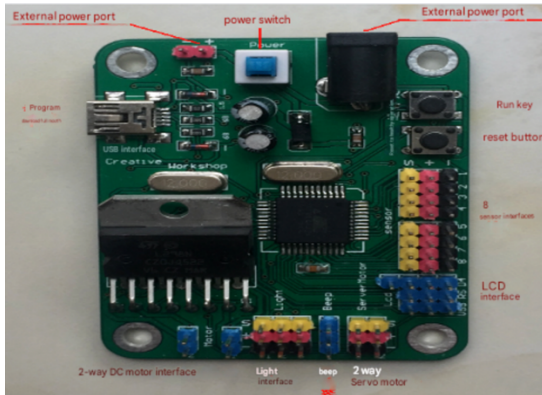


Figure 2 Controller

3.3 Sensor Selection and Decision

Compared to traditional mechanical buttons, capacitive touch-sensitive buttons are not only aesthetically pleasing but also durable and long-lasting. They revolutionize traditional mechanical button control by enabling on/off control, quantitative adjustment, and even directional control with a simple touch. Capacitive touch-sensitive buttons are now widely used in consumer products such as mobile phones, DVD players, and washing machines.

An equivalent capacitance is formed between any two adjacent plates in a circuit. Therefore, in capacitive sensing applications, these two electrodes are artificially created to obtain an equivalent capacitance. The image above shows a capacitive sensing demonstration board. It features seven rectangular copper foils surrounded by ground, creating an equivalent capacitance between the foils and ground. Each copper foil is connected to an IO port on the sensing chip via a wire, resulting in an equivalent static capacitance between

each IO port and ground. In practice, these copper foils act as physical buttons, sensing changes in external capacitance. As shown in the diagram below, when no finger is pressed, each button has an equivalent static capacitance to ground. When a finger is pressed, the human body capacitance is added to the button, increasing the capacitance between the button and ground. The resulting capacitance, C_p , is then input into the chip for processing.

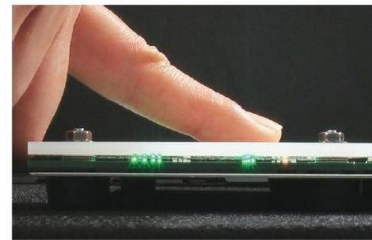


Figure 3 Principle of Sensor Operation

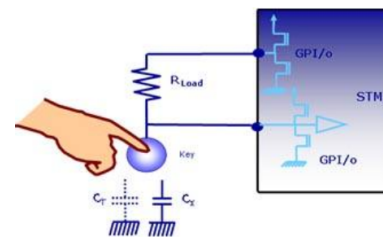


Figure 4 Principle of Sensor Operation

3.4 IoT Module Design

This module can be controlled remotely via the Internet or short-range Bluetooth, while providing instant feedback on local device status to the phone.

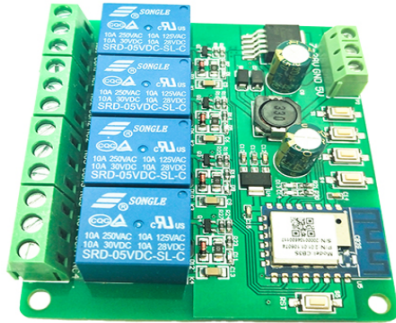


Figure 5

3.5 Mechanical Structure

We sketched a design based on the existing elevator buttons, then used CAD to create the design on the computer. Finally, we used laser engraving.

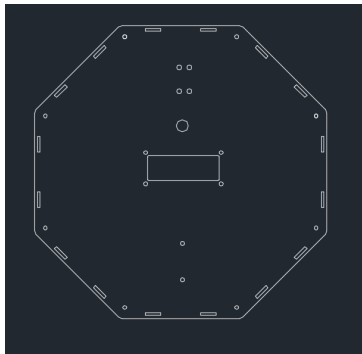


Figure 6 Structure Drawings(1)

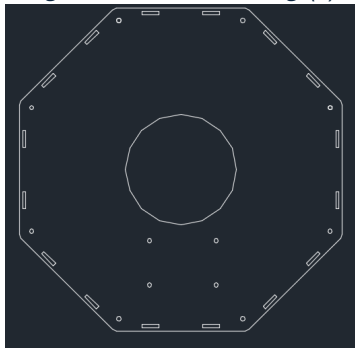


Figure 7 Structure Drawings(2)

Drawings of the upper and lower panels, with holes reserved for the LCD, buzzer, and indicator lights. I chose acrylic as the processing material, as it is stronger than wood and less susceptible to moisture. Below are drawings of the mechanical arm and motor coupling.

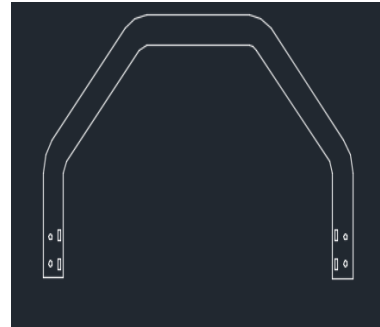


Figure 8 Lock fence drawings(1)

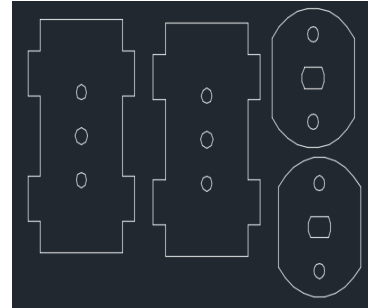


Figure 9 Lock fence drawings(2)



Figure 10 Final product(1)



Figure 11 Final product(2)

3.6 Electronic Circuits

After consulting relevant materials, we installed our motor and wires onto our structure. This part of the process presented significant challenges: the overall aesthetics were compromised, and the tangled wiring

caused wires to fall out, making it difficult to identify the ports.

We ultimately used a glue gun to glue each wire connector together to prevent them from falling out, then used cable ties to secure the wires. The controller and other components were then installed and secured inside the ground lock. This aesthetically pleasing design didn't affect the device's operation, resulting in excellent results.

The image below shows the devices surrounding the controller. Input devices are shown in orange, and output devices are shown in light blue.

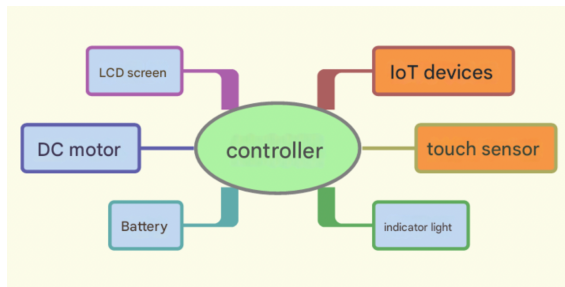


Figure 12

3.7 Software Programming

By writing programs using graphical programming software, it's intuitive and allows you to realize your ideas. Then, download them to the controller for testing, ultimately achieving your vision in reality.

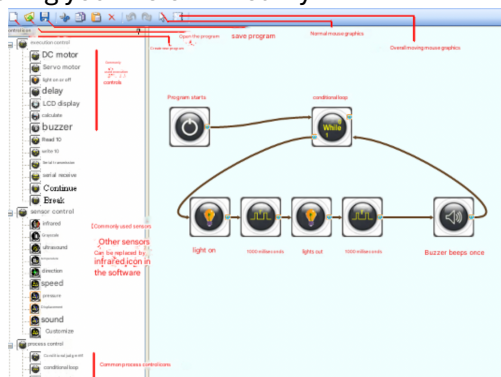


Figure 13

3.8 Program design

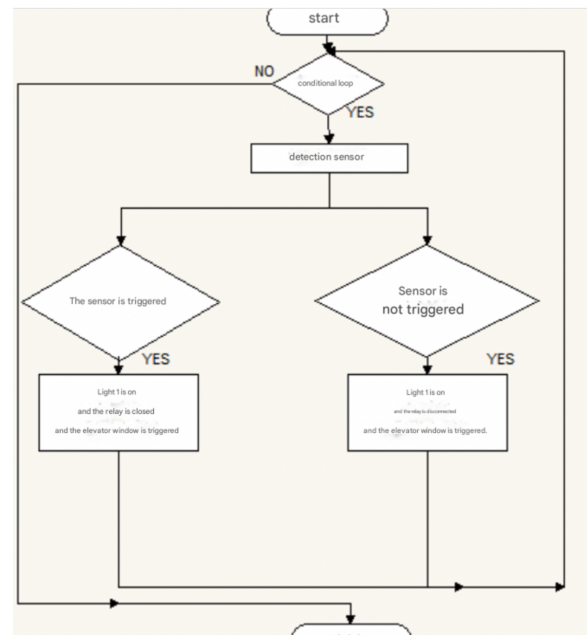


Figure 14 Planning program flow chart

4. Project Description

Smart Floor Lock. The owner can use their phone's Bluetooth to control the floor lock's opening and closing. If the parking space is unused for an extended period, the owner can share the space using their phone. The floor lock's screen will display the owner's phone number, indicating that the space has been shared. This is convenient for those who have not used the space for an extended period. Someone can call the number on the LCD screen to inquire about the time and cost of sharing the space. The owner will then provide a password. Entering the password unlocks the floor lock.

A contactless touch sensor captures the password information. Password entry uses a single-touch numeric input method, minimizing the number of keystrokes required. For example, if the password is 5, touch a button five times in succession, then touch

another sensor to confirm the 5-digit password. If the number of accidental touches exceeds 5, for example 6, simply continue touching the sensor until the number reaches 9, at which point the number will reset and resume counting from 0. All progress is displayed on the LCD screen.

Our chosen wireless control method uses an internet-connected module, enabling both remote and short-range Bluetooth control. The phone displays the ground lock's status in real time and allows for real-time control. For example, to open or close the ground lock, a corresponding graphical interface appears, which can be tapped. To share a parking space, simply tap the share graphical interface on your phone. If you make a mistake, you can simply tap the reset graphical interface on your phone to reset the entire ground lock system.



Figure 15 Test

5. Conclusion

Sensor Selection: I previously considered a photoelectric sensor, but it was susceptible to light interference. Furthermore, the sensor's sensing area is exposed to the outside, making it susceptible to dust contamination. Ultimately, I opted for a non-contact touch sensor. This sensor can accommodate wooden boards, plastic, or ceramic lamps.

Wireless Module Selection: I initially selected a proximity sensor with a wireless remote control range of 50 meters in open space. However, this module was significantly susceptible to interference in the complex environment of the garage. Given the actual use case, I needed remote control. Therefore, we abandoned this module and opted for a Bluetooth proximity sensor that offers both remote control and Bluetooth proximity control.

6. References

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